Global Horizon Bank

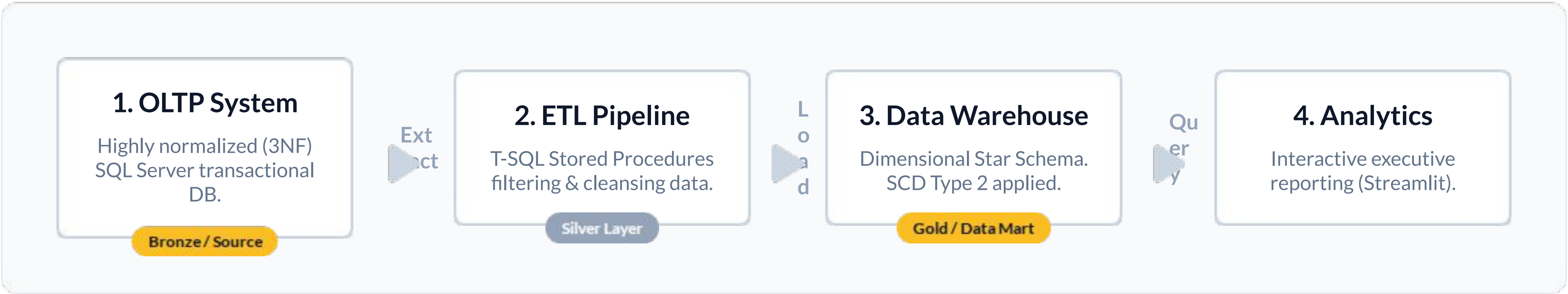
Data Architecture

End-to-End Enterprise Data Pipeline (OLTP to OLAP)

Applying Medallion Architecture & Data Lineage

Enterprise Data Pipeline Overview

The high-level data flow representing the extraction, transformation, and loading of banking data into analytical structures.



Phase 1: Source to Bronze (OLTP Schema)

Extracting Raw Transactions

Data is ingested from the core operational database as-is. The structure is highly normalized (3NF) to support transactional integrity, not analytical querying.

- **Relational Complexity:** Requires complex joins across Customers, Accounts, and Transactions.
- **Raw State:** Contains raw strings and operational IDs without historical tracking.

Customers
CustomerID PK
FirstName
LastName
DateOfBirth

Accounts
AccountID PK
CustomerID FK
AccountType
Balance

Transactions
TransactionID PK
AccountID FK
TransactionType
Amount
TransactionDate

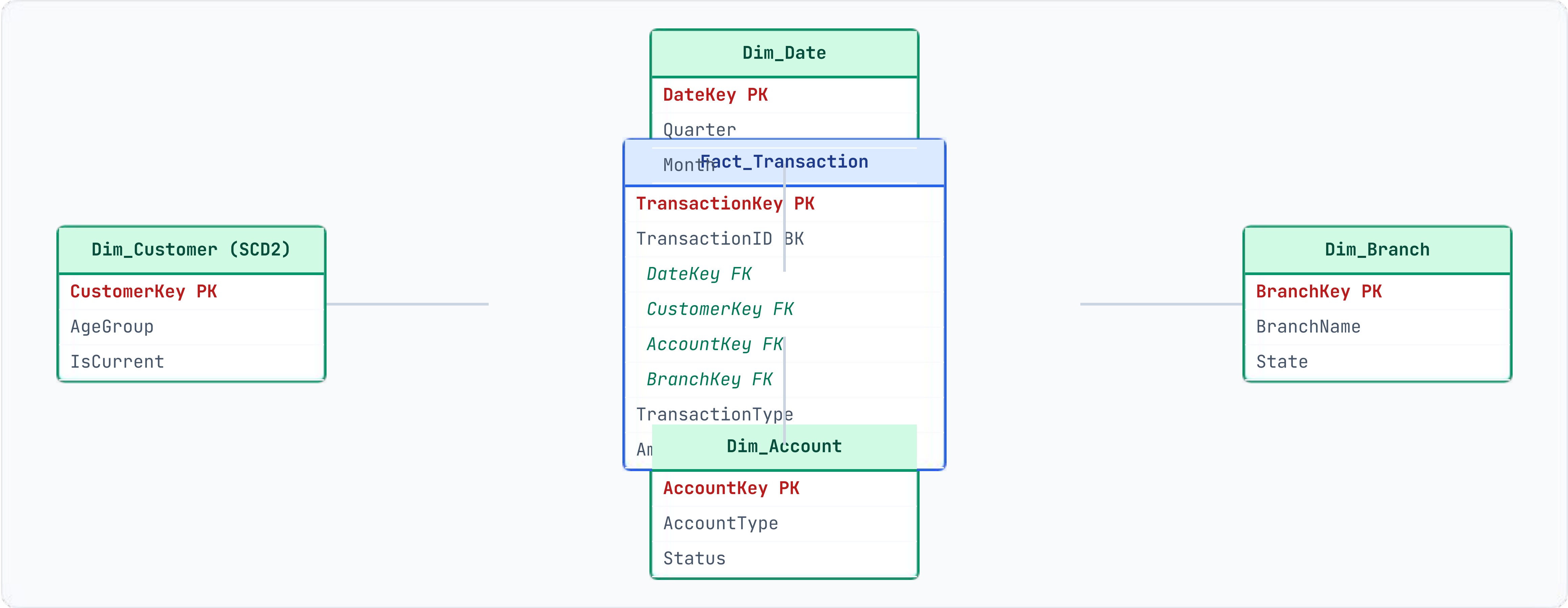
Data Lineage: Transformation in Silver

Mapping the evolution of attributes from the normalized OLTP source to the denormalized OLAP targets through the ETL process.

OLTP Source Field (Bronze)	ETL Transformation Logic (Silver)	Data Mart Target (Gold)
Customers.**CustomerID**	→ Generate Surrogate Key, Apply SCD2 logic	Dim_Customer.**CustomerKey** (PK)
Transactions.**TransactionDate**	→ Parse, Extract temporal components (Qtr, Day)	Dim_Date.**DateKey** (FK)
Transactions.**Amount**	→ Cleanse nulls, cast to fixed decimal	Fact_Transaction.**Amount**
Customers.**DateOfBirth**	→ Calculate age dynamically, bucket into ranges	Dim_Customer.**AgeGroup**

Phase 3: The Gold Layer (OLAP Star Schema)

The heavily transformed data is exposed as a Star Schema, optimizing read performance and analytical queries for the dashboard.



Pipeline Complete

Data structure is successfully optimized for dashboard ingestion.